PHD125 Pharmaceutical Organic Chemistry · Amino Acids, Proteins and Enzymes

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Amino Acids, Proteins and Enzymes CH7-LO1

The α-carbon of an amino acid carries:

A. $-NH_2$, –COOH, –H and a side chain

B. two –COOH groups

C. a phosphate

D. a benzene ring always

A2 Topic: Amino Acids, Proteins and Enzymes CH7-LO1

Which amino acid has a side chain that forms disulfide bridges?

A. cysteine

B. serine

C. alanine

D. lysine

A3 Topic: Amino Acids, Proteins and Enzymes CH7-LO1

Amino acids in human proteins are:

A. L
B. D
C. racemic
D. achiral

A4 Topic: Amino Acids, Proteins and Enzymes CH7-LO2

At a pH above its pI, an amino acid carries a net:

A. negative charge
B. positive charge
C. zero charge
D. double positive charge

A5 Topic: Amino Acids, Proteins and Enzymes CH7-LO2

Which amino acid has the highest pI?

A. arginine
B. aspartic acid
C. alanine
D. serine

A6 Topic: Amino Acids, Proteins and Enzymes CH7-LO3

The bond joining amino acids in a protein is a(n):

A. amide (peptide) bond
B. ester bond
C. glycosidic bond
D. disulfide bond

A7 Topic: Amino Acids, Proteins and Enzymes CH7-LO3

How many different dipeptides can be made from one alanine and one glycine?

A. 2
B. 1
C. 4
D. 3

A8 Topic: Amino Acids, Proteins and Enzymes CH7-LO4

β-Pleated sheets are held together by hydrogen bonds:

A. between neighbouring chain segments
B. within one turn of a helix
C. between side chains only
D. with water only

A9 Topic: Amino Acids, Proteins and Enzymes CH7-LO4

Heavy metal ions such as Hg^{2+} denature proteins by binding:

A. –SH and $–COO^-$ groups
B. peptide bonds
C. sugars
D. phosphates

A10 Topic: Amino Acids, Proteins and Enzymes CH7-LO5

An enzyme that hydrolyses fats is a:

A. lipase
B. kinase
C. isomerase
D. ligase

A11 Topic: Amino Acids, Proteins and Enzymes CH7-LO5

Above about 50 °C, most enzyme reactions slow down because the enzyme:

A. denatures
B. becomes saturated
C. is hydrolysed to amino acids
D. turns into a cofactor

A12 Topic: Amino Acids, Proteins and Enzymes CH7-LO5

Most coenzymes are made from:

A. water-soluble vitamins
B. fatty acids
C. cholesterol
D. glucose

Part B · Structured questions

Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Amino Acids, Proteins and Enzymes CH7-LO1 7 marks, C2

(a) Draw the general structure of an α-amino acid and label the α-carbon and the two functional groups. (2 marks)

(b) Classify serine, leucine, aspartic acid and lysine according to their side chains. (2 marks)

(c) Explain why glycine is not chiral but alanine is. (1 mark)

(d) State which form, D or L, is found in human proteins, and name two essential amino acids. (2 marks)

Question B2 Topic: Amino Acids, Proteins and Enzymes CH7-LO2 8 marks, C3

(a) Explain what a zwitterion is, and use it to explain why alanine is a high-melting solid that dissolves in water. (2 marks)

(b) Draw the main form of alanine at pH 1, pH 6 and pH 12, and give the net charge of each. (3 marks)

(c) Define the isoelectric point, and predict the direction of movement of aspartic acid (pI 3.0), alanine (pI 6.0) and lysine (pI 9.8) during electrophoresis at pH 6. (3 marks)

Question B3 Topic: Amino Acids, Proteins and Enzymes CH7-LO3 7 marks, C3

(a) Write an equation, using structural formulas, for the formation of the dipeptide Gly-Ser. Circle the peptide bond and name the type of reaction. (3 marks)

(b) Explain why Gly-Ser and Ser-Gly are different compounds, and identify the N-terminal amino acid in each. (2 marks)

(c) Explain how a single change in amino acid sequence causes sickle cell anaemia. (2 marks)

Question B4 Topic: Amino Acids, Proteins and Enzymes · CH7-LO4 · 8 marks, C3

(a) Describe the four levels of protein structure, stating the bonds or interactions that hold each. (4 marks)

(b) Distinguish between protein hydrolysis and denaturation. (2 marks)

(c) Explain why (2 marks)

- **(i)** 70% ethanol is used to disinfect skin, and
- **(ii)** insulin must be stored at 2–8 °C.

Question B5 Topic: Amino Acids, Proteins and Enzymes · CH7-LO5 · 8 marks, C4

(a) Explain what the active site of an enzyme is, and compare the lock-and-key and induced-fit models. (2 marks)

(b) Explain why the rate of an enzyme reaction levels off at high substrate concentration, and why it falls above about 50 °C. (2 marks)

(c) Atorvastatin is a competitive inhibitor of HMG-CoA reductase. Explain what this means and what happens if the concentration of the natural substrate rises. (2 marks)

(d) Aspirin irreversibly inhibits cyclooxygenase (COX) in platelets. Explain why one low dose affects platelets for several days. (2 marks)

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Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Amino Acids, Proteins and Enzymes CH7-LO1

The α-carbon of an amino acid carries:

A. $-NH_2$, –COOH, –H and a side chain

B. two –COOH groups

C. a phosphate

D. a benzene ring always

FGOB 18.3.

A2 Topic: Amino Acids, Proteins and Enzymes CH7-LO1

Which amino acid has a side chain that forms disulfide bridges?

A. cysteine

B. serine

C. alanine

D. lysine

$-CH_2SH$.

A3 Topic: Amino Acids, Proteins and Enzymes CH7-LO1

Amino acids in human proteins are:

A. L

B. D

C. racemic

D. achiral

Except glycine, which is not chiral.

A4 Topic: Amino Acids, Proteins and Enzymes CH7-LO2

At a pH above its pI, an amino acid carries a net:

A. negative charge

B. positive charge

C. zero charge

D. double positive charge

It has lost H^+.

A5 Topic: Amino Acids, Proteins and Enzymes CH7-LO2

Which amino acid has the highest pI?

A. arginine

B. aspartic acid

C. alanine

D. serine

Basic side chain: pI 10.8.

A6 Topic: Amino Acids, Proteins and Enzymes CH7-LO3

The bond joining amino acids in a protein is a(n):

A. amide (peptide) bond

B. ester bond

C. glycosidic bond

D. disulfide bond

–CO–NH–.

A7 Topic: Amino Acids, Proteins and Enzymes CH7-LO3

How many different dipeptides can be made from one alanine and one glycine?

A. 2

B. 1

C. 4

D. 3

Ala-Gly and Gly-Ala.

A8 Topic: Amino Acids, Proteins and Enzymes CH7-LO4

β-Pleated sheets are held together by hydrogen bonds:

A. between neighbouring chain segments

B. within one turn of a helix

C. between side chains only

D. with water only

FGOB 18.7.

A9 Topic: Amino Acids, Proteins and Enzymes CH7-LO4

Heavy metal ions such as Hg^{2+} denature proteins by binding:

A. –SH and $–COO^-$ groups

B. peptide bonds

C. sugars

D. phosphates

FGOB 18.10.

A10 Topic: Amino Acids, Proteins and Enzymes CH7-LO5

An enzyme that hydrolyses fats is a:

A. lipase

B. kinase

C. isomerase

D. ligase

A hydrolase.

A11 Topic: Amino Acids, Proteins and Enzymes CH7-LO5

Above about 50 °C, most enzyme reactions slow down because the enzyme:

A. denatures

B. becomes saturated

C. is hydrolysed to amino acids

D. turns into a cofactor

FGOB 19.5.

A12 Topic: Amino Acids, Proteins and Enzymes CH7-LO5

Most coenzymes are made from:

A. water-soluble vitamins

B. fatty acids

C. cholesterol

D. glucose

FGOB 19.2.

Part B · Structured questions

Question B1 Topic: Amino Acids, Proteins and Enzymes CH7-LO1 7 marks, C2

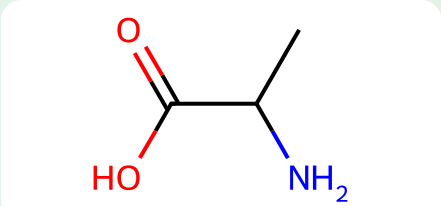
(a) Draw the general structure of an α-amino acid and label the α-carbon and the two functional groups. (2 marks)

(b) Classify serine, leucine, aspartic acid and lysine according to their side chains. (2 marks)

(c) Explain why glycine is not chiral but alanine is. (1 mark)

(d) State which form, D or L, is found in human proteins, and name two essential amino acids. (2 marks)

(a) $H_2N–CH(R)–COOH$: the α-carbon carries the amino group ($–NH_2$), the carboxyl group (–COOH), H and the side chain R.



(b) Serine: polar, uncharged ($–CH_2OH$). Leucine: nonpolar. Aspartic acid: acidic. Lysine: basic.

(c) In glycine R = H, so the α-carbon has two identical H atoms; in alanine it carries four different groups (H, CH_3, NH_2, COOH).

(d) L. Any two of His, Ile, Leu, Lys, Met, Phe, Thr, Trp, Val.

Step	Marking criteria	Marks
1	Correct general structure	1
2	α-carbon, amino and carboxyl groups labelled	1
3	Serine polar; leucine nonpolar	1
4	Aspartic acid acidic; lysine basic	1
5	Glycine has two H on the α-carbon	1
6	L form	1
7	Two correct essential amino acids	1
	Total	**7**

Question B2 Topic: Amino Acids, Proteins and Enzymes CH7-LO2 8 marks, C3

(a) Explain what a zwitterion is, and use it to explain why alanine is a high-melting solid that dissolves in water. (2 marks)

(b) Draw the main form of alanine at pH 1, pH 6 and pH 12, and give the net charge of each. (3 marks)

(c) Define the isoelectric point, and predict the direction of movement of aspartic acid (pI 3.0), alanine (pI 6.0) and lysine (pI 9.8) during electrophoresis at pH 6. (3 marks)

(a) A zwitterion has a + charge ($-NH_3^+$) and a − charge ($-COO^-$) on the same molecule, formed when the acid gives H^+ to its own amine. Strong ionic attractions between zwitterions give a high melting point, and the charges interact well with water, like a salt.

(b) PH 1: $H_3N^+–CH(CH_3)–COOH$, +1. pH 6: $H_3N^+–CH(CH_3)–COO^-$, 0. pH 12: $H_2N–CH(CH_3)–COO^-$, −1.

(c) The pI is the pH at which the amino acid is a zwitterion with no net charge. At pH 6: aspartic acid (negative) moves to the positive electrode; alanine (neutral) does not move; lysine (positive) moves to the negative electrode.

Step	Marking criteria	Marks
1	Zwitterion defined (+ and − on one molecule)	1
2	Ionic attractions → high mp; salt-like → soluble	1
3	pH 1 form, +1	1
4	pH 6 zwitterion, 0	1
5	pH 12 form, −1	1
6	pI defined	1
7	Asp → + electrode; Ala does not move	1
8	Lys → − electrode	1
	Total	**8**

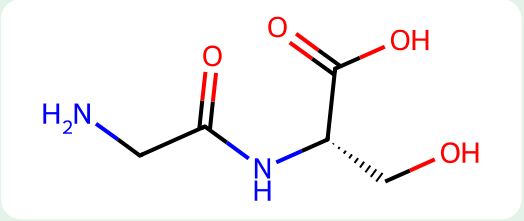
Question B3 Topic: Amino Acids, Proteins and Enzymes CH7-LO3 7 marks, C3

(a) Write an equation, using structural formulas, for the formation of the dipeptide Gly-Ser. Circle the peptide bond and name the type of reaction. (3 marks)

(b) Explain why Gly-Ser and Ser-Gly are different compounds, and identify the N-terminal amino acid in each. (2 marks)

(c) Explain how a single change in amino acid sequence causes sickle cell anaemia. (2 marks)

(a) $H_2N–CH_2–COOH + H_2N–CH(CH_2OH)–COOH \rightarrow H_2N–CH_2–$**CO–NH–**$CH(CH_2OH)–COOH + H_2O$. The CO–NH is the peptide (amide) bond; it is a condensation reaction.



(b) The peptide bond joins different groups: in Gly-Ser glycine's –COOH is joined to serine's $–NH_2$; in Ser-Gly it is the other way round, so the structures differ. N-terminal: glycine in Gly-Ser; serine in Ser-Gly.

(c) In the haemoglobin β chain, valine (nonpolar) replaces glutamic acid (charged). The haemoglobin molecules stick together into long fibres, distorting red blood cells into a sickle shape that blocks small vessels.

Step	Marking criteria	Marks
1	Correct reactants and products with water	1
2	Peptide bond identified	1
3	Condensation reaction	1
4	Different groups joined → different compounds	1
5	Correct N-terminals	1
6	Val replaces Glu in haemoglobin	1
7	Haemoglobin aggregates; sickled cells	1
	Total	**7**

Question B4 Topic: Amino Acids, Proteins and Enzymes CH7-LO4 8 marks, C3

(a) Describe the four levels of protein structure, stating the bonds or interactions that hold each. (4 marks)

(b) Distinguish between protein hydrolysis and denaturation. (2 marks)

(c) Explain why (2 marks)

- **(i)** 70% ethanol is used to disinfect skin, and
- **(ii)** insulin must be stored at 2–8 °C.

(a) Primary: the amino acid sequence, held by peptide bonds. Secondary: α-helix and β-sheet, held by hydrogen bonds between backbone C=O and N–H. Tertiary: the overall fold of one chain, held by side-chain ionic bonds, hydrogen bonds, disulfide bridges and hydrophobic/dispersion interactions. Quaternary: the arrangement of two or more chains, held by the same side-chain interactions (e.g. haemoglobin).

(b) Hydrolysis breaks peptide bonds (needs strong acid or proteases) and gives amino acids; denaturation unfolds the 2°, 3° and 4° structure but leaves the peptide bonds (primary structure) intact.

(c) **(i)** Ethanol denatures microbial proteins by disrupting their hydrogen bonds.

(ii) Heat (or freezing) can denature insulin, a protein, so it would stop working.

Step	Marking criteria	Marks
1	Primary: sequence, peptide bonds	1
2	Secondary: helix/sheet, backbone H-bonds	1
3	Tertiary: side-chain interactions (named)	1
4	Quaternary: several chains	1
5	Hydrolysis breaks peptide bonds → amino acids	1
6	Denaturation unfolds; primary structure kept	1
7	Ethanol denatures microbial proteins	1
8	Heat/freezing denatures insulin	1
	Total	**8**

Question B5 Topic: Amino Acids, Proteins and Enzymes CH7-LO5 8 marks, C4

(a) Explain what the active site of an enzyme is, and compare the lock-and-key and induced-fit models. (2 marks)

(b) Explain why the rate of an enzyme reaction levels off at high substrate concentration, and why it falls above about 50 °C. (2 marks)

(c) Atorvastatin is a competitive inhibitor of HMG-CoA reductase. Explain what this means and what happens if the concentration of the natural substrate rises. (2 marks)

(d) Aspirin irreversibly inhibits cyclooxygenase (COX) in platelets. Explain why one low dose affects platelets for several days. (2 marks)

(a) The active site is the pocket on the enzyme where the substrate binds and reacts. Lock and key: a rigid site that exactly fits the substrate. Induced fit: the site changes shape when the substrate binds, to fit it closely.

(b) At high [S] all active sites are occupied (the enzyme is saturated), so adding substrate cannot increase the rate. Above about 50 °C the enzyme denatures and the active site loses its shape.

(c) Atorvastatin resembles the substrate and binds reversibly to the active site, competing with the substrate; if [substrate] rises, more substrate wins the site and the rate recovers towards normal.

(d) Aspirin forms a covalent bond (acetylates a serine) in COX, permanently inactivating it; platelets have no nucleus and cannot make new COX, so the effect lasts for the platelet's life (days), until new platelets are made.

Step	Marking criteria	Marks
1	Active site defined	1
2	Lock-and-key vs induced fit	1
3	Saturation of active sites	1
4	Denaturation above ~50 °C	1
5	Competitive: resembles substrate, binds active site	1
6	More substrate overcomes the inhibition	1
7	Covalent, permanent inactivation	1
8	Platelets cannot make new COX	1

	Total	**8**

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